

Executive Summary: Liver Fibrosis Biomarker Discovery

Prepared for Karyon Bio

Project Objective

This project applied a multi-omics single-cell RNA-seq pipeline to discover and prioritize cell-type-specific biomarkers and therapeutic targets for human liver fibrosis/cirrhosis, a major cause of death worldwide with no effective antifibrotic therapies. The analysis integrates transcriptomic evidence from single-cell data with human genetic evidence to identify the highest-confidence targets for diagnostic and therapeutic development.

Dataset

GSE136103 (Ramachandran et al., Nature 2019): 55,341 cells from 10 human liver donors (5 healthy, 5 cirrhotic), 10X Chromium scRNA-seq, 20 FACS-sorted fractions. This is the landmark study that defined the fibrotic niche at single-cell resolution.

Key Findings

- **23 cell types identified**, including all 5 disease-relevant populations from the original publication: TREM2+CD9+ scar-associated macrophages, ACKR1+/PLVAP+ scar-associated endothelial cells, and PDGFRa+ collagen-producing myofibroblasts.
- **856 differentially expressed genes** across 20 cell types (Cirrhotic vs Healthy), with disease-relevant compartments showing the most changes (Kupffer cells: 303 DEGs, sinusoidal endothelial: 162, scar-associated endothelial: 139, scar-associated macrophages: 88, myofibroblasts: 61).
- **186 enriched pathways**, including metabolic reprogramming in myofibroblasts (oxidative phosphorylation, glycolysis, hypoxia) and suppressed inflammatory signaling in Kupffer cells (TNF-alpha/NF-kB).
- **84,352 ligand-receptor interactions** mapped, confirming key fibrotic niche pathways: JAG1-NOTCH, DLL4-NOTCH1, SPP1-CD44, and TNF-TNFRSF1A signaling.
- **160 targets with human genetic evidence** for liver cirrhosis (Open Targets GWAS/ClinVar), including SERPINA1, PNPLA3, TNFRSF1A, and NFKB1.
- **6 targets with convergent** genetic and transcriptomic evidence (highest confidence).

Top 10 Ranked Biomarker/Target Candidates

Rank	Gene	Score	Tier	Key Evidence	Translational Angle
1	SERPINA1	0.541	HIGH	DE in 1 CTs, GWAS=0.91, L-R	Both
2	NFKB1	0.522	HIGH	DE in 1 CTs, GWAS=0.84	Both
3	C3	0.494	HIGH	DE in 4 CTs, L-R	Diagnostic
4	KRT18	0.471	HIGH	DE in 2 CTs, GWAS=0.87	Both
5	VWF	0.442	HIGH	DE in 7 CTs, L-R	Diagnostic
6	SPP1	0.441	HIGH	DE in 3 CTs, L-R	Diagnostic
7	FN1	0.425	HIGH	DE in 4 CTs, L-R	Diagnostic
8	TIMP1	0.392	HIGH	DE in 1 CTs, L-R	Diagnostic

9	CD9	0.388	HIGH	DE in 1 CTs, L-R	Diagnostic
10	KRT8	0.374	HIGH	DE in 1 CTs, GWAS=0.83	Both

Translational Recommendations

- **Highest-priority therapeutic targets:** SERPINA1 (convergent genetic+transcriptomic, alpha-1 antitrypsin deficiency is a known cirrhosis cause), NFKB1 (central inflammatory TF, genetically validated), and TNFRSF1A (TNF receptor, anti-TNF biologics already exist for other indications).
- **Best diagnostic biomarker candidates:** SPP1 (secreted osteopontin, DE in 3 cell types), C3 (complement, DE in 4 cell types), VWF (endothelial marker, DE in 7 cell types), and CD9 (surface marker of scar-associated macrophages). These are accessible in serum or via flow cytometry.
- **Recommended validation experiments:** (1) IHC for TREM2, CD9, ACKR1, PLVAP to confirm scar-restricted localization; (2) flow cytometry for TREM2+CD9+ macrophage expansion in independent cohorts; (3) serum ELISA for SPP1, C3, SERPINA1; (4) functional assays for NFKB1/TNFRSF1A inhibition in fibrotic co-culture models.

Next Steps

- Validate top targets in independent liver cirrhosis cohorts (e.g., TCGA-LIHC, GTEx liver) - Perform spatial transcriptomics to confirm scar-restricted expression of top candidates - Conduct druggability assessment and compound screening for NFKB1, TNFRSF1A, C3 pathway - Develop serum biomarker panel (SPP1, C3, SERPINA1, VWF) for non-invasive fibrosis staging

Data and Code Availability

All code, results, and figures are available in the reproducible project package (liver_fibrosis_biomarker_pipeline.zip). The analysis is built on GSE136103 from GEO. Random seed: 42. Software versions documented in requirements.txt.